ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 3RD YEAR

SOFTWARE ENGINEERING

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BSc Computer Science, 3rd year, semester 1 final examination: software process models and Agile, requirements engineering, design and UML, design principles and patterns, software testing, project planning and maintenance.

Written by Academic Central for practice, following third-year Computer Science syllabuses at South African universities (software engineering). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SIX questions. Answer ALL the questions.
2. Read the case study before you answer QUESTION 3.
3. Draw diagrams neatly in pen or pencil, using standard UML notation.
4. Calculators may be used.
5. Number your answers exactly as the questions are numbered.

SECTION A

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 Which process model finishes each phase completely before the next phase starts?

A Waterfall
B Scrum
C Spiral
D Extreme Programming

1.2 In Scrum, who orders the product backlog so that the most valuable work is done first?

A Scrum Master
B Product Owner
C Developers
D Project sponsor

1.3 "A search must return results within 2 seconds" is an example of a …

A functional requirement.
B non-functional requirement.
C user story.
D test case.

1.4 Which UML diagram best shows the order in which objects send messages to each other over time?

A Class diagram
B Use case diagram
C Sequence diagram
D Deployment diagram

1.5 A well-designed system has modules with …

A high coupling and low cohesion.
B low coupling and high cohesion.
C high coupling and high cohesion.
D low coupling and low cohesion.

1.6 Testing done by the client to decide whether to take over the finished system is called …

A unit testing.
B integration testing.
C acceptance testing.
D stress testing.

1.7 Running earlier tests again after a change, to check that nothing that worked before is now broken, is called …

A regression testing.
B smoke testing.
C load testing.
D alpha testing.

1.8 Changing software so that it runs on a new version of the operating system is …

A corrective maintenance.
B adaptive maintenance.
C perfective maintenance.
D preventive maintenance.

1.9 Which design pattern makes sure that a class has only one instance?

A Observer
B Factory Method
C Singleton
D Adapter

1.10 Which Git command records the staged changes as a new version in the local repository?

A git push
B git commit
C git clone
D git pull

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SECTION B

QUESTION 2: SOFTWARE PROCESSES

2.1 Describe the waterfall model. Give TWO kinds of projects that it suits and ONE weakness of the model. (4)

2.2 State the FOUR values of the Agile Manifesto. (4)

2.3 Explain what a sprint is in Scrum, and give the purpose of each of the four Scrum events that take place in a sprint. (5)

2.4 Explain why the spiral model suits large projects with high risk. (2)

[15]

QUESTION 3: REQUIREMENTS ENGINEERING

CASE STUDY: CampusEats. A university wants a mobile app, CampusEats, for its food kiosks. Students log in with their student number, browse the menus of the kiosks, place an order and pay from their student-card balance through the university's payment system. Kiosk staff see new orders on a tablet and mark each order as "ready"; the student then gets a notification to collect the food. An administrator adds kiosks and updates the menus and prices. About 20 000 students use the campus, and most orders are placed between 12:00 and 14:00.

3.1 Name THREE stakeholders of CampusEats and give ONE interest (need) of each. (3)

3.2 Write FOUR functional requirements and TWO non-functional requirements for CampusEats. For each non-functional requirement, name its type (for example performance or security). (6)

3.3 Write ONE user story for a student collecting an order, in the standard user-story form, with TWO acceptance criteria in Given–When–Then form. (4)

3.4 Draw a UML use case diagram for CampusEats. Show the actors Student, Kiosk staff, Administrator and Payment system, at least FIVE use cases, the system boundary and at least ONE «include» relationship. (5)

3.5 Name TWO techniques for eliciting (gathering) requirements. (2)

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QUESTION 4: DESIGN

4.1 Draw a UML class diagram for the description below. Show the classes with their attributes, the associations with multiplicities, and the generalisation. (8)

A course has a code, a name and a number of credits. A lecturer has a staff number and a name. Each course is taught by exactly one lecturer, and a lecturer teaches one or more courses. A student has a student number and a name. A student can enrol in many courses and a course has many students; for each enrolment the date and the final mark are stored. Undergraduate students and postgraduate students are kinds of students. Each postgraduate student has one supervisor, who is a lecturer.

4.2 Explain, with an example, why low coupling makes software easier to maintain. (2)

4.3 A weather station must tell a phone app and a web dashboard every time it takes a new reading. More kinds of displays will be added later.

4.3.1 Name the design pattern that suits this problem. (1)

4.3.2 Describe the roles of the classes in this pattern and how the displays get the new reading. (3)

4.3.3 Give ONE benefit of using the pattern here. (1)

4.4 Name and briefly explain the FIVE SOLID principles of object-oriented design. (5)

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QUESTION 5: SOFTWARE TESTING

5.1 A function ticket_price(age) takes a whole number age. Ages from 0 to 12 pay R20, ages from 13 to 59 pay R50, and ages from 60 to 120 pay R30. Any other age is invalid and gives an error.

5.1.1 Identify ALL the equivalence partitions (valid and invalid) and give ONE test value for each. (5)

5.1.2 List the boundary values that you would test. (3)

5.2 Study the function below.

```
def grade(mark):
    if mark < 0 or mark > 100:
        return "invalid"
    if mark >= 75:
        result = "distinction"
    elif mark >= 50:
        result = "pass"
    else:
        result = "fail"
    return result
```

5.2.1 Calculate the cyclomatic complexity of grade. Show how you got your answer. (3)

5.2.2 Give the smallest set of input values that gives 100% branch coverage, and show which branch each value covers. (3)

5.3 Distinguish between black-box and white-box testing. (2)

5.4 Explain unit testing, integration testing and system testing. (3)

5.5 Name the three steps of the test-driven development (TDD) cycle, in order. (1)

[20]

QUESTION 6: PROJECT MANAGEMENT AND MAINTENANCE

6.1 The table shows the activities of a small software project.

Activity	Description	Duration (days)	Depends on
A	Requirements	3	–
B	Database design	4	A

C	User interface design	5	A
D	Back-end coding	6	B
E	Front-end coding	4	C
F	Integration testing	3	D, E
G	Deployment	1	F

6.1.1 Draw the activity network (activity-on-node). (3)

6.1.2 Find the critical path and the shortest time in which the project can be finished. (3)

6.1.3 Calculate the slack (float) of activity E. (2)

6.2 Explain the difference between corrective and perfective maintenance, with an example of each. (2)

6.3 Give TWO benefits of using version control such as Git in a team, and explain what a merge conflict is. (3)

6.4 Name TWO risks to a software project and give a way to reduce each. (2)

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TOTAL: 100

END OF QUESTION PAPER